

卷四: Discrete Mathematics—Final Test

考试 注意 事项	一、答题纸为空白 A4 纸; 二、考试结束后, 老师通知结束答题。请将答案图片粘到 word, 导出为 pdf, 发往邮箱 bupt_2021@163.com, 邮件主题: 离散-期末-姓名-学号。pdf 名: 离散-期末-姓名-学号.pdf。 三、学生作答试卷不得抄袭, 如被发现, 按相应规定严肃处理。
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考试课程	离散数学													
题号	一	二	三	四	五	六	七	八	九	十	十一	十二	十三	总分
满分	8	8	8	6	8	8	8	8	8	8	8	6	8	
得分														
阅卷教师														

1. [8 points] For each of these relations on the set $\{1, 2, 3, 4\}$, decide whether it is reflexive, whether it is symmetric, whether it is antisymmetric, and whether it is transitive. Please answer in the form with $\checkmark$ and $\times$.

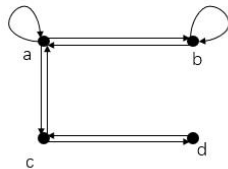
- a) $\{(2, 3), (3, 2)\}$
 b) $\{(1, 2), (2, 3), (3, 4), (2, 4)\}$
 c) $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$
 d) $\{(1, 3), (1, 4), (1, 1), (2, 4), (3, 1), (3, 4)\}$

	reflexive	symmetric	antisymmetric	transitive
a)				
b)				
c)				
d)				

2. [8 points] Determine whether the relations represented by these zero–one matrices or the relation with the directed graph are equivalence relations. Determine the properties of an equivalence relation that the others lack.

a)
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

b)
$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$



c)

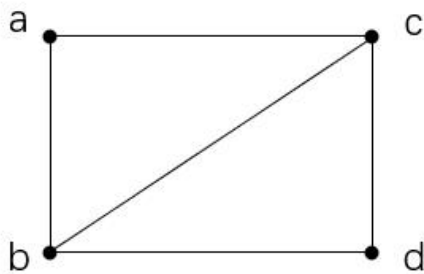


d)

3. [8 points] Suppose $A = \{2, 3, 5, 6, 7, 10, 12, 14, 15, 21\}$ and R is the partial order relation defined on A where xRy means x is a divisor of y .
- Draw the Hasse diagram for R .
 - Find all maximal elements.
 - Find all minimal elements.
 - Find $\text{lub}(\{3, 7\})$, if it exists.
 - Find $\text{glb}(\{12, 15\})$, if it exists.
 - Is the $\text{poset}(A, R)$ a lattice? Explain your answer.

4. [6 points] Use Warshall's algorithm to find the transitive closure of R on $\{a, b, c, d, e\}$ where $R = \{(a, b), (a, c), (a, e), (b, a), (b, c), (c, a), (c, b), (d, a), (e, d)\}$.

5. [8 points] Answer the following questions about chromatic number.
- Find the chromatic number of C_n , where $n \geq 3$.
 - Find the chromatic polynomial P_G for the given graph and use P_G to find $\chi(G)$.



6. [8 points] Answer the following questions about recurrence relation.
- Solve the recurrence relation $a_n = 8a_{n-1} - 16a_{n-2}$ for $n \geq 2$ with the initial condition $a_0 = 0$ and $a_1 = -4$.
 - Use generating functions to solve the recurrence relation $a_n = 3a_{n-1}$ for $n \geq 1$ with the initial condition $a_0 = 5$.

7. [8 points] Let H and N be subgroups of the group G . Prove that if N is a normal subgroup of G , then $H \cap N$ is a normal subgroup of H .

Proof:

Step 1: $H \cap N$ is a subgroup of H .

To prove $H \cap N$ is a subgroup of H , we check the three subgroup criteria for $H \cap N$:

1. **Identity:**

- Since N is a subgroup of G , it contains the identity element e .
- Since e is also in H , $e \in H \cap N$.
- Thus, $H \cap N$ contains the identity element.

2. **Closure:**

- Let $x, y \in H \cap N$. Then $x, y \in H$ and $x, y \in N$.
- Since H and N are subgroups, their elements are closed under the group operation. Thus, $xy \in H$ and $xy \in N$, so $xy \in H \cap N$.
- Hence, $H \cap N$ is closed under the group operation.

3. **Inverses:**

- Let $x \in H \cap N$. Then $x \in H$ and $x \in N$.
- Since H and N are subgroups, $x^{-1} \in H$ and $x^{-1} \in N$, so $x^{-1} \in H \cap N$.
- Hence, $H \cap N$ is closed under inverses.

Thus, $H \cap N$ is a subgroup of H .

Step 2: $H \cap N$ is normal in H .

To show that $H \cap N$ is normal in H , we need to prove that for all $h \in H$ and all $x \in H \cap N$,

$$h x h^{-1} \in H \cap N.$$

1. $h x h^{-1} \in H$:

- Since $h, x \in H$ and H is a subgroup, $h x h^{-1} \in H$.

2. $h x h^{-1} \in N$:

- Since $x \in H \cap N$, $x \in N$.
- Since N is normal in G , for any $h \in G$, $h x h^{-1} \in N$.
- In particular, since $h \in H \subseteq G$, $h x h^{-1} \in N$.

3. **Conclusion:**

- Since $h x h^{-1} \in H$ and $h x h^{-1} \in N$, we conclude that $h x h^{-1} \in H \cap N$.

Thus, $H \cap N$ is normal in H .

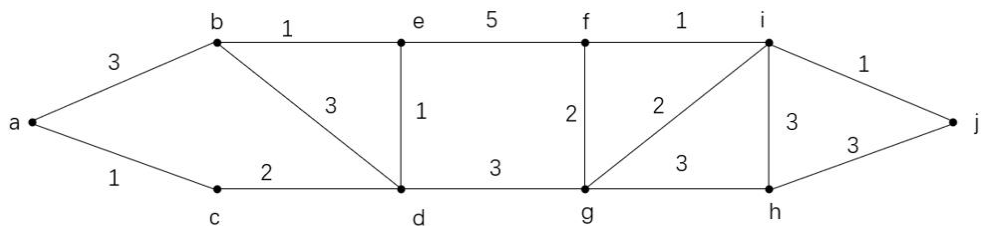
8. [8 points] Answer the following questions about group.
- Write the multiplication table for the group $Z_2 \times Z_3$.
 - $\langle Z_6, +_6 \rangle$ is a group, $Z_6 = \{[0], [1], [2], [3], [4], [5]\}$, determine all the subgroup of $\langle Z_6, +_6 \rangle$.
 - Let $G = Z_4$. Determine all the left cosets of $H = \{[0], [2]\}$ in G .

9. [8 points] Let

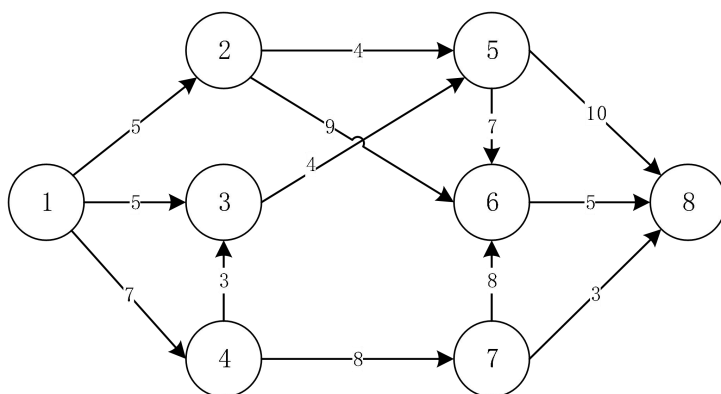
$$H = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

be a parity check matrix. Decode the following words relative to a maximum likelihood decoding function associated with e_H .

- Find all left cosets of $e_H(B^2)$ in B^5 .
 - For each coset, find a coset leader, and compute the syndrome of each leader.
 - Decode the word 10101 relative to a maximum likelihood decoding function.
10. [8 points] Use Dijkstra's algorithm to find the length of a shortest path from a to all vertices in the given weighted graph.



11. [8 points] Find the maximal flow for the network N given in following figure.



12. [6 points] Use Huffman coding to encode the following symbols with the frequencies listed: A: 0.08, B: 0.10, C: 0.14, D: 0.15, E: 0.18, F: 0.35.

- a) Write down the prefix code of these symbols.
b) What is the average number of bits used to encode a character?

13. [8 points] Use Prim's algorithm to find a minimum spanning tree for the given weighted graph.

